

UNIT 2 · INTRODUCTION AND STATE OF ART OF AI, NLP & POTENTIAL USE OF AI

Chapter 2.1

Brief History and Elements of AI

Machine Learning · Deep Learning · Applications in Image, Speech & Information Retrieval



Learning Objectives

1

Understand Machine Learning

Build a clear concept of what ML is and how it differs from conventional programming.

3

Learn the types of ML

Distinguish Supervised, Unsupervised, and Reinforcement Learning.

5

Study Image & Speech Recognition

Learn how machines see and listen — step by step.

2

Explore ML applications

Survey real-world use cases across industries and everyday technology.

4

Understand Deep Learning

See how neural networks extend machine learning with layered architectures.

6

Grasp Information Retrieval (IR)

Understand how search engines find and rank relevant results.

Two definitions worth remembering

“

Machine learning is a field of study that gives computers the ability to learn without being explicitly programmed.

— Arthur Samuel, 1959

“

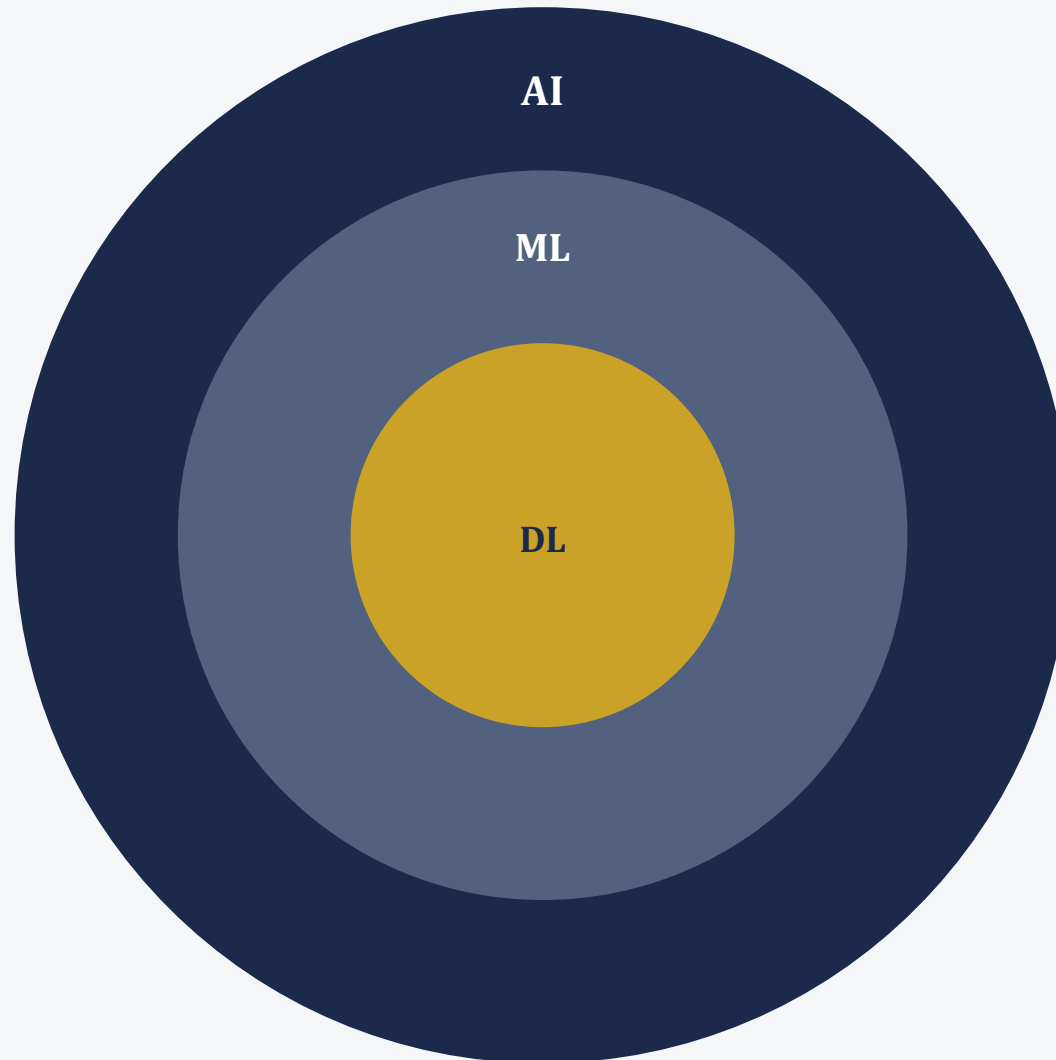
Machine learning is the science of getting computers to act without being explicitly programmed.

— Stanford University

The core idea

Instead of writing step-by-step code, we feed data to a generic algorithm — and the algorithm builds its own logic from that data. ML is a subset of AI, and Deep Learning is a subset of ML.

AI, Machine Learning & Deep Learning



Artificial Intelligence

The broadest field — systems that simulate human-like intelligence and decision-making.



Machine Learning

A specialised branch of AI that learns patterns automatically from data.



Deep Learning

A subset of ML using layered neural networks to learn features directly from raw data.

Learning Pasta the Way a Machine Learns

1

Try & Notice

First attempt: too little salt, pasta overcooked.
You notice the results.



2

Find the Pattern

After a few tries, you realise 8–10 minutes
and a teaspoon of salt works best.



3

Apply Automatically

Next time, you follow the learned steps
without checking the recipe.

Key Insight

Just like Machine Learning, you improve performance over time by learning from past outcomes and adjusting future actions.

Conventional Programming vs. Machine Learning

Example: Celsius → Fahrenheit conversion

Conventional Programming

- 1 Input: Celsius value
- 2 Programmer writes the rule:
 $F = C \times 1.8 + 32$
- 3 Computer applies the rule
- 4 Output: Fahrenheit value

Machine Learning Approach

- 1 Feed many Celsius values:
-40, 10, 0, 8, 15, 22, 38
- 2 Feed matching Fahrenheit values:
-40, 14, 32, 46, 59, 72, 100
- 3 Pass both sets to the ML algorithm
- 4 Algorithm predicts F for any new C

Three Types of Machine Learning

1

Supervised Learning

Learns from labelled data — like a student guided by a teacher.

$$Y = f(X)$$

2

Unsupervised Learning

Finds hidden patterns in unlabelled data with no explicit guidance.

Find structure

3

Reinforcement Learning

Learns by trial and error — rewarded for good actions, penalised for bad.

Reward / Penalty

Learning From Labelled Examples

Like a student learning from an instructor — the student already knows the correct answers while learning from the course.

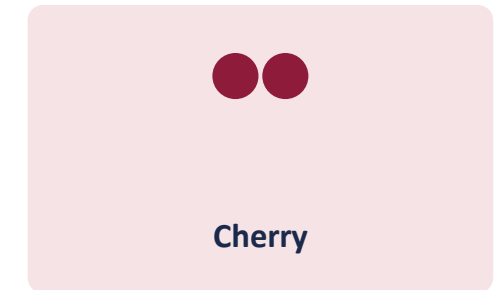
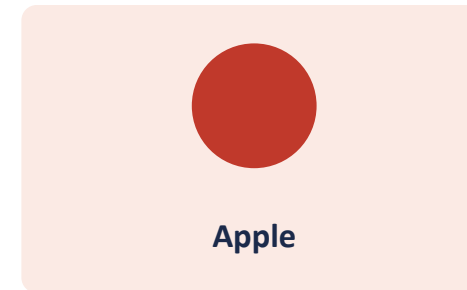
Common Algorithms

- Nearest Neighbor
- Decision Trees
- Linear Regression
- Neural Networks

$$Y = f(X)$$

Approximate the mapping function so well that for new input X , the model predicts output Y .

Example: Apple or Cherry?

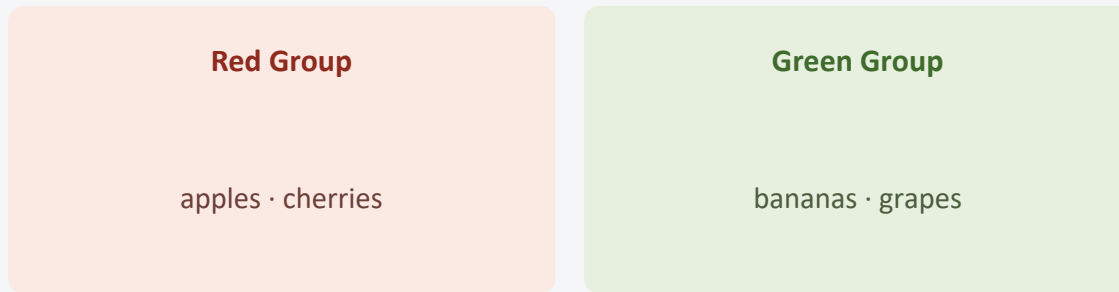


1. Train the model with many labelled images of apples and cherries.
2. Model learns the visual features that separate the two classes.
3. Shown a new, unlabelled image, it predicts the correct label.

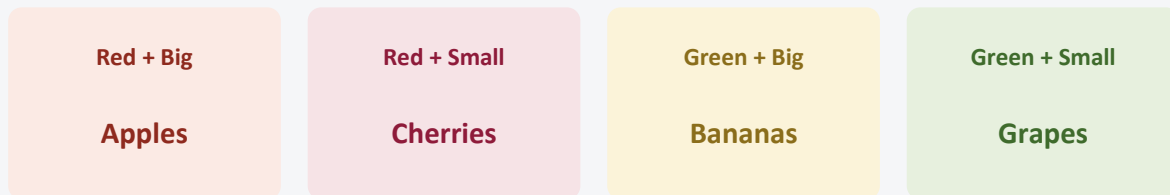
Discovering Structure Without Labels

No prior knowledge of fruit types or labels — the system organises a mixed basket using only natural characteristics.

Step 1 — Group by Color



Step 2 — Refine by Size



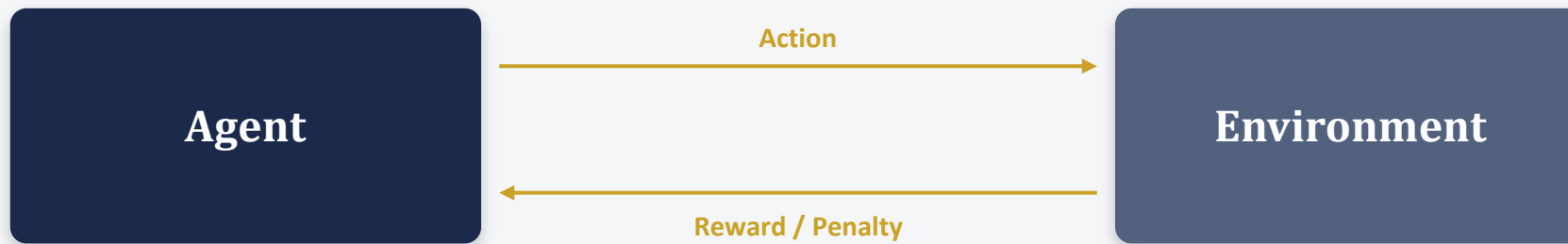
Insight

This process reflects unsupervised learning: the system automatically discovers hidden patterns and structures — here, color and size — without ever being told what the categories are.

Characteristics

- Automatically extracts features and finds patterns
- Uncovers hidden structures in data
- More difficult to implement than supervised learning

Learning Through Trial and Reward



Positive action → positive feedback / reward. **Negative action** → penalty. The agent learns automatically from this feedback loop — without labelled data — improving performance through experience.

Applications

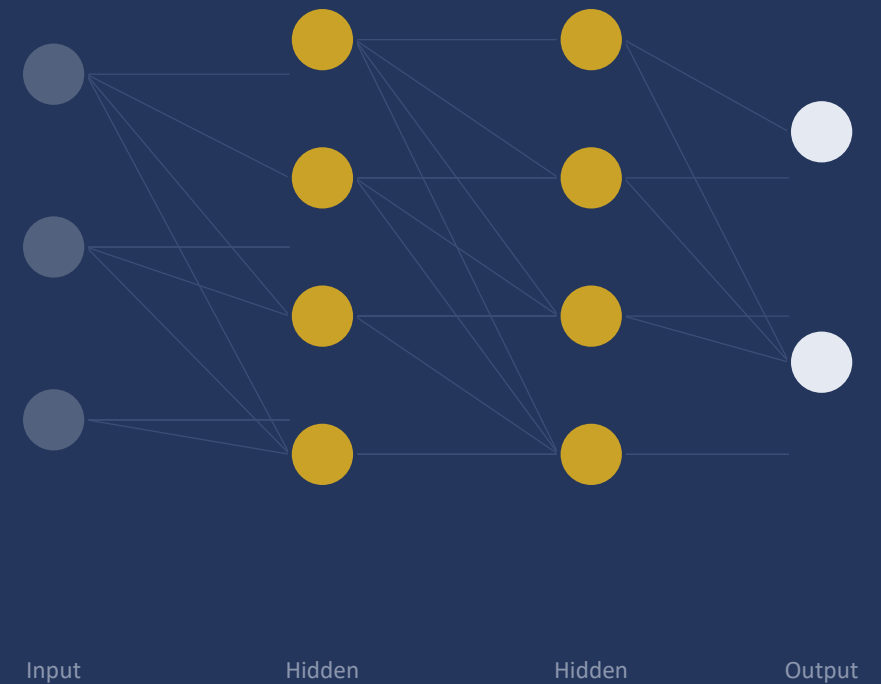
- Video Games — complete levels, earn badges/bonuses
- Board Games — Chess, Go (computer-played)
- Robotics & Self-Driving Cars — learn safe actions

What is Deep Learning?

Deep Learning teaches computers to do what comes naturally to humans, using software that mimics the network of neurons in the brain — learning directly from vast amounts of data.

- Learning is done through a neural network — layers stacked on top of each other.
- Depth = number of layers the model uses to learn from data.
- Trained on large sets of labelled data — features are learned automatically, no manual feature extraction.
- Considered the current state of the art in AI.

A Layered Neural Network



Machine Learning vs. Deep Learning

Aspect	Machine Learning	Deep Learning
Data	Works with small / medium datasets	Needs very large datasets
Hardware	Runs on normal computers	Requires GPU / TPU
Features	Manual feature extraction	Learns features automatically
Speed	Faster training (minutes–hours)	Slower (hours–weeks)
Algorithms	Many (SVM, DT, KNN)	Few (CNN, RNN, GANs)
Interpretability	Transparent, easy to explain	Black-box, hard to explain
Use Cases	Spam filter, fraud detection	Image, speech, NLP, driving

KEY APPLICATION AREAS

Where ML & DL Meet the Real World

1

Image Recognition

Teaching computers to see and classify what's in a picture.

2

Speech Recognition

Turning spoken words into understood, actionable commands.

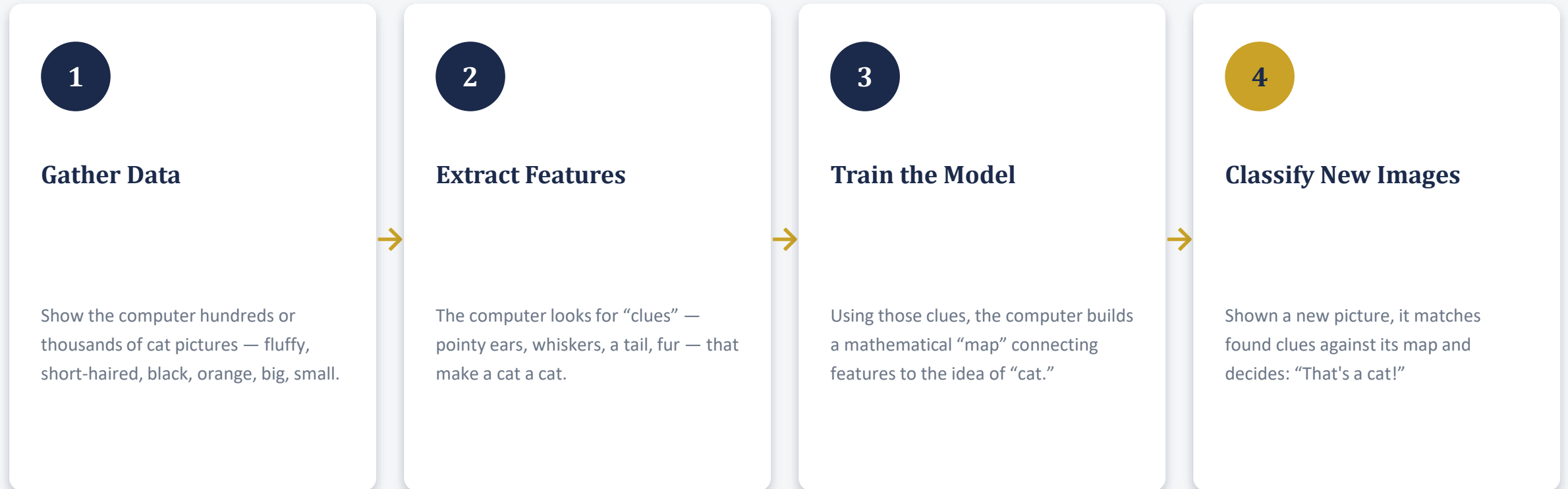
3

Information Retrieval

Finding the most relevant results from vast amounts of data.

How Image Recognition Works

A walkthrough using the classic example: teaching a computer to recognise a cat.



How Speech Recognition Works

The Five-Step Process

- 1 Sound Waves — your voice creates waves picked up by a microphone
- 2 Digitisation — sound waves are converted into digital data
- 3 Break Down Sounds — split into phonemes / small word parts
- 4 Match to Words — compared against a huge dictionary of sounds
- 5 Understand Meaning — grammar & context reveal what you meant

Key Components

Acoustic Model — the computer's “ear”; links sounds to word parts

Language Model — the “brain” for grammar and word order

Pronunciation — handles accents and speech variations

Context — resolves sound-alikes like “hear” vs “here”

Example: Ordering Pizza

“I'd like to order a large pepperoni pizza with extra cheese.”

→ Sound digitised → broken into parts → matched to words → meaning understood
→ order placed.

How Search Engines Use Information Retrieval

Like a super-smart librarian: finding the most relevant information from millions of documents.

1

Understand Query

Breaks the query into keywords, identifies intent, considers context.

2

Query Processing

Compares the query to its index of web pages for relevant keywords.

3

Rank Results

Scores by relevance, popularity, quality, and personalisation.

4

Present Results

Displays the most relevant results in a clear, user-friendly format.

Searching Effectively

- Use specific keywords
- Use “quotes” for exact phrases
- Use AND / OR / NOT operators
- Always evaluate source credibility

Key Terms to Remember

● Machine Learning

Systems that automatically learn and improve from experience without explicit programming.

● Supervised Learning

Classifies/predicts using labelled training data.

● Unsupervised Learning

Discovers patterns in data with no labels or guidance.

● Reinforcement Learning

Learns via rewards and penalties from actions taken.

● Neural Networks

Brain-inspired mathematical models that store learned information.

● Deep Learning

Uses large labelled datasets and layered neural networks to learn features automatically.

“Machine learning algorithms can figure out how to perform important tasks by generalizing from examples.”

— University of Washington

Thank You

Questions & Discussion



Dr. Saptarshi Jana

saptarshi.jana@techvein.com